

K702 — Are we still missing the K-type positions? YES for 3 of 4 sectors — but the finish is now well-posed: each sector is a SPINOR TOWER (E-ladder ground + conformal climbing); charged leptons done; the up FOLLOWS from the down; the last real derivations are the DOWN-quark and NEUTRINO E-ladder grounds.

Keeper | 2026-07-15 | Casey: “are we still missing the k-type positions, can we finish that?” The FK weight is resolved ($E_0=2$ by spin, K701). The charged-lepton positions are pinned. What remains: the OTHER three sectors’ positions. This is finishable — well-posed, not open-ended — and here’s the exact path.

What we HAVE

- The weight: $E_0=2$, forced by spin (Elie 4680, Lyra confirming) — no longer a citation-wall. The mixing uses the CONFORMAL-ENERGY (E-ladder / climbing) positions, weighted by $E_0=2$.
- The charged-lepton E-ladder: $E = 2, 3, 4$ ($E_0=2$ = the 5D free-fermion dimension, +1 per conformal raise). Positions $\{1, 2/3, 1/2\}$. Done.

What we’re MISSING (the honest answer: 3 of 4 sectors)

The mixing needs the E-ladder (climbing/conformal-energy) positions for all four sectors. Only the charged leptons are pinned. Missing: - Up quarks — but these FOLLOW from the down (F548: the up = the down refracted by the index $3/2$; small refraction → small CKM). Derivable once the down is pinned — NOT an independent unknown. - Down quarks — the “true ground,” colored ($N_c=3$), bulk. Its E-ladder GROUND E_{0_down} is the remaining derivation. - Neutrinos — a “different tower,” chargeless (the $\nu=1/2$ address is on the ν -parameter axis, NOT the E-ladder axis — K689 two-axes caution). Its E-ladder GROUND E_{0_nu} is the remaining derivation. (The big E-gap between the charged-lepton tower at 2,3,4 and the neutrino tower → large PMNS.)

THE FINISH PATH (well-posed — two grounds, then the up follows)

Each sector is a spinor tower: a ground E_0 (the spinor boundary weight for that sector’s field content) + conformal climbing (+1 per raise). The charged-lepton ground $E_0=2$ came from the spinor weight (free-fermion dim, colorless/boundary). So: 1. Derive the DOWN-quark E-ladder ground E_{0_down} — from its field content: colored ($N_c=3$), bulk-localized, at the $\nu=N_c$ address (F506). The SAME spinor-weight logic that gave the leptons $E_0=2$, applied to a colored bulk fermion. (The down positions for MIXING are the E-ladder, not the mass-ladder $\{1,3,5\}$ — Lyra’s positions≠masses caution.) 2. The UP follows — the down refracted by the index $3/2$ (F548). No independent derivation. 3. Derive the NEUTRINO E-ladder ground E_{0_nu} — from its field content: chargeless, the different tower. (The $\nu=1/2$ address is the ν -parameter, not directly the E-ladder position — pin the E-ladder value for the chargeless fermion.) 4. Then Grace runs the six — the four sectors’ E-ladder positions → $U_charged^\dagger U_nu$ (PMNS), $U_u^\dagger U_d$ (CKM) → all six + θ_{23} octant.

CAN WE FINISH IT? — Yes, and it’s ~2 derivations, not open-ended

- NOT open-ended: the framework (spinor tower per sector, E_0 + climbing) is in hand; the weight is forced ($E_0=2$); the up follows from the down. The remaining work is two E-ladder grounds (down, neutrino), each a well-posed rep-theory derivation from the sector’s field content (color, charge) by the same logic that gave the leptons $E_0=2$.
- The discipline (Elie/Lyra held it): don’t fabricate positions you can’t force. So the two grounds must be DERIVED (from color/charge), not fitted to the angles. That’s the honest last step.
- Keeper’s structural read (for the team, not a guess at the values): the leptons (colorless, boundary) ground at $E_0=2 = (d-1)/2$. The quarks are colored/bulk → their ground likely shifts by the color/bulk structure; the neutrinos are chargeless/different-tower → their ground sits low (giving the large PMNS gap). Deriving

those two shifts from the field content is the finish — Lyra+Elie's lane. Once the down and neutrino grounds land, the table renders.

The 26 — state

14 banked (α closed), 6 identified-strong, 6 behind the two remaining E-ladder grounds (down + neutrino) — the up and the run follow. This is the last derivation: two sector grounds, then Grace's one-pass run.

— Keeper K702, 2026-07-15. Missing: the up/down/neutrino E-ladder positions (charged leptons done, $E=2,3,4$). Finishable + well-posed: each sector = spinor tower (E_0 + climbing); the up FOLLOWS from the down ($3/2$ refraction); the two real derivations are the DOWN and NEUTRINO E-ladder grounds (from color/charge field content, same logic as leptons' $E_0=2$). Then Grace runs the six. Don't fabricate — derive the grounds. See [[Keeper_K701...]], [[Keeper_K689...]] (two-axes: E-ladder vs ν -parameter).